ITM University Gwalior

Database management system (CSLO311)

Synopsis

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AI-Assisted Blood Bank and Donor Matching System

Submitted To

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1. Introduction

Blood is one of the most important resources in healthcare, especially during emergencies, surgeries and other critical medical situations. A blood bank must manage several activities at the same time, such as maintaining donor records, recording donations, testing blood samples, tracking blood units, making sure that blood isn’t expired, blood is stored in a clean surrounding its needed and fulfilling hospital requests.

These activities are handled manually or through disconnected systems, it can become difficult to maintain accurate records and quickly find the required blood unit. There can also be problems such as using expired blood being delays in emergency allocation and difficulty in checking donor eligibility or maybe blood cannot provide to the patient at the correct time.

The **AI-Assisted Blood Bank and Donor Matching System** is proposed as a database-based solution for managing these activities in a structured and reliable way. The system focuses on maintaining relationships between donors, donations, blood units, laboratory tests, hospitals and allocation records. It also uses database-level rules to improve data consistency and safety.

The project follows a relational database approach with PostgreSQL as the proposed database platform. The database is designed using normalization principles and includes transaction handling, access control, audit records and AI-assisted features for future inventory prediction.

2. Problem Statement

Blood banks need to maintain accurate information about donors, blood units, testing and hospital requirements. In an inefficient system, tracking the expiry of blood units can result in unnecessary wastage, while delays in matching and allocating blood can become serious during emergencies.

Another important issue is donor eligibility. A donor should not be allowed to donate again before the required interval has passed. If this information is maintained manually, there is a possibility of missing or incorrectly entering previous donation details.

There is also a need for proper tracking of the movement and status of blood units. A fragmented record of custody makes it harder to investigate problems, perform audits or trace a blood unit when required.

Therefore, the project aims to design a centralized and reliable database system that can manage blood-bank operations, enforce important business rules, support quick donor and blood-unit matching, and maintain proper records for auditing and future analysis.

3. Need of Proposed System

The system can help in:

- Maintaining a centralized record of donors and donations.
- Tracking individual blood units and their expiry dates.
- Recording laboratory screening and testing results.
- Checking donor eligibility before accepting a new donation.
- Reducing delays in finding suitable blood units for hospitals.
- Preventing the same blood unit from being allocated to multiple requests.

4. Objectives

The main objectives of the project are:

1. To design a relational database for managing blood-bank operations.
2. To maintain complete records of donors, donations, blood units, tests and hospitals.
3. To apply normalization principles and reduce data redundancy.
4. To implement primary keys, foreign keys, constraints and other integrity rules.
5. To enforce donor eligibility and the mandatory donation interval through database rules.
6. To track blood-unit availability, allocation, expiry and quarantine status.
7. To support faster and more reliable blood matching for hospital requests.
8. To handle emergency allocation using proper transaction and concurrency control.
9. To maintain audit records for important changes in blood-unit status.To provide an AI-ready database structure for predicting shortages and improving donor outreach.

5. Scope of the Project

The scope of the project covers the database side of blood-bank management.

The system mainly deals with:

- Donor registration and donor history.
- Donation records.
- Blood-unit generation and tracking.
- Laboratory screening and test records.
- Blood storage locations.
- Hospital profiles and blood requests.

- Blood-unit allocation and transfer.
- Cold-chain and expiry information.
- Transfusion records.
- Audit logs and user roles.
- AI-assisted analysis of historical blood-supply data.

The project is primarily focused on designing and implementing the database and its supporting operations. The AI component is intended to work with historical and sanitized data to provide predictions such as possible shortages and suggested donor outreach.

6. Proposed System

The proposed system uses a centralized relational database in which different blood-bank activities are connected through well-defined relationships.

A donor can have multiple donation records, and a donation can produce one or more blood components or units. Each unit is associated with information such as blood group, collection time, expiry time and storage location. Laboratory tests are linked with the corresponding donation or blood components.

Hospitals can submit blood requests based on their requirements. The allocation module connects hospital requests with suitable blood units. This relationship is important because one hospital may require multiple units and the available units may be considered for different requests.

The database also maintains information about technicians, inventory locations, cold-chain events, transfusions, audit logs and user roles. This gives the system a complete view of the movement and status of blood units. The proposed model contains more than ten major entities to cover operational, safety and auditing requirements.

7. Database Design

The database design is based on the following major entities:

- Donor
- Donation
- Blood Unit
- Laboratory Test
- Hospital
- Blood Request
- Allocation
- Technician
- Inventory Location
- Cold Chain Event
- Transfusion

Some of the important relationships are:

Donor → Donation

One donor can have multiple donation records.

Donation → Blood Unit

One donation can result in one or more blood components or units.

Blood Unit ↔ Hospital Request

A blood unit can be considered for different requests, while a request may require multiple blood units. This many-to-many relationship is handled using the Allocation entity.

Donation → Tests

Testing information is maintained against the corresponding donation or blood component.

These relationships allow the database to trace information from a donor and donation to the eventual allocation of a blood unit.

8. Normalization

Normalization is an important part of the proposed database design because blood-bank records contain several related types of information.

The database is designed up to **BCNF (Boyce-Codd Normal Form)** where applicable. Repeating and multi-valued information is separated into different relations to reduce redundancy. Functional dependencies and candidate keys are considered while designing the relations.

For example, information related to blood-unit storage and expiry is maintained with the blood-unit record instead of repeatedly storing the same information in hospital or donation records.

Primary keys and foreign keys are used to maintain relationships between tables. Referential integrity rules are also applied so that important records are not accidentally removed or left without their related records.

9. Important Business Rules

The system includes database-level rules for important blood-bank operations.

Donor Eligibility

A donor must satisfy the required eligibility conditions before a new donation is accepted. The system also checks the mandatory **90-day interval** from the previous donation.

Blood Unit Status

A blood unit is not treated as available until the required screening tests have been completed successfully. Units can have statuses such as:

- Available

- Allocated
- Expired
- Quarantined

Expiry and Cold Chain

Blood units affected by expiry or cold-chain problems are removed from the matching process and can be marked as expired or quarantined.

Allocation

During an emergency, the system considers factors such as blood-group compatibility, age of the blood unit and proximity while selecting an appropriate unit.

10. Integrity Constraints

Different database constraints are used to prevent invalid data from entering the system.

Some examples include:

- Primary key constraints for unique identification of records.
- Foreign key constraints for maintaining relationships.
- CHECK constraints for valid blood groups and statuses.
- Date validation for donation records.
- Unique constraints for donor identification where applicable.
- Triggers for enforcing donation-interval rules.
- Functions and procedures for important database operations.

For example, the database can reject a new donation if the donor's previous donation was less than 90 days ago. This prevents the rule from depending only on the application interface.

11. Transactions and Concurrency Control

Emergency blood allocation can involve multiple hospitals requesting the same blood unit at almost the same time. Without proper concurrency control, the same unit could accidentally be allocated to more than one request.

To avoid this situation, the proposed system uses database transactions and row-level locking.

For example, when selecting the last available O-negative unit, the system locks the selected row before changing its status to allocated. If the unit is successfully reserved, the transaction is committed. If no suitable unit is available or an error occurs, the transaction can be rolled back.

This approach uses ACID properties to maintain consistency during important operations. The project specifically proposes SELECT ... FOR UPDATE with transaction control to prevent double allocation during simultaneous requests.

12. AI-Assisted Features

The database is designed to support AI-assisted analysis in addition to normal database operations.

Historical blood-supply and usage information can be used to identify patterns and forecast possible shortages based on blood group and location. These predictions can help in planning future inventory requirements and donor outreach.

The AI component is kept separate from the main transactional database. Sanitized and anonymized data can be provided to analytical models so that the main audit and operational records remain authoritative.

The PBL guidelines require AI to be used responsibly and state that AI-generated work should be manually reviewed and validated before being used in the project.

13. Security and Audit

Since blood-bank information is sensitive, the system requires controlled access to database information.

Role-Based Access Control (RBAC) is proposed to separate permissions for different users such as:

- Clinical staff
- Laboratory staff
- Administrators
- Auditors

An audit-log mechanism is also included to record important changes made to blood-unit records. The audit record can contain the actor, timestamp, previous state, new state and location of the blood unit.

This makes it easier to trace changes and supports auditing and investigation when required.

14. Technology Used

The proposed technical stack includes:

Component	Technology
Database	PostgreSQL
Database Testing	pgTAP
Schema Migration	Liquibase / Flyway
Version Control	Git
Database Design	ER Model and Relational Schema
AI/Analytics	ML-based predictive analysis
Deployment	Cloud database / managed database as applicable

The PBL instructions also recommend maintaining the database as code, using version-controlled schema and migration scripts, automated testing and appropriate deployment practices.

15. Expected Outcomes

After implementing the proposed system, the project is expected to provide:

- A structured database for blood-bank management.
- Reduced duplication and inconsistency in records.
- Better tracking of blood units and expiry dates.
- Automated enforcement of important donor eligibility rules.
- Faster identification and allocation of suitable blood units.

- Prevention of duplicate allocation during concurrent requests.
- Improved traceability through audit records.
- Controlled access through role-based security.
- A database structure that can support future AI-based prediction and analytics.

The overall aim is to make blood-bank data management more organized and reliable while keeping the database ready for further improvements.

16. Future Scope

The system can be extended in several ways in the future.

1. **Mobile/Web Application:** A user-friendly application can be developed for donors, hospitals and blood-bank staff.
2. **Real-Time Notifications:** Hospitals and donors can receive alerts for urgent requirements, donation eligibility and blood shortages.
3. **Improved AI Prediction:** More historical data can be used to improve shortage forecasting.
4. **Donor Recommendation:** The system can suggest suitable donors based on blood group, location and eligibility.
5. **Cloud Deployment:** The complete database can be deployed on a managed cloud database for better availability and scalability.
6. **Hospital Integration:** The system can be integrated with hospital information systems for faster blood requests and allocation.
7. **Advanced Analytics:** Dashboards can be developed to monitor blood demand, wastage, expiry trends and regional shortages.

The PBL guidelines also identify deployment, monitoring, backup/recovery and future evolution as important parts of a professionally engineered database system.

17. Conclusion

The **AI-Assisted Blood Bank and Donor Matching System** is designed to address common problems in blood-bank data management by bringing donor records, blood units, testing, hospital requests and allocation information into one relational database system.

The project applies important DBMS concepts such as ER modeling, normalization, constraints, triggers, transactions, concurrency control, indexing, role-based access and audit logging. It also provides a foundation for using AI on historical data to predict blood shortages and support better inventory planning.

By combining database principles with a real-world healthcare problem, the project aims to provide a system that is not only useful for storing information but also capable of maintaining consistency, safety and traceability during important blood-bank operations.

18. References

1. Database Management Systems (CSL0311), DBMS Problem-Based and Project-Based Learning (PbBL) Guidelines, ITM University Gwalior.
2. PostgreSQL Documentation.
3. Project Presentation: **AI-Assisted Blood Bank & Donor Matching System**.
4. Documentation and references related to database normalization, transactions, concurrency control and role-based access control.